Data Science & Business Analytics

Full Program — Course Syllabus

Master the complete analytics-to-machine-learning pipeline with Python, SQL, statistics, BI dashboards, predictive modeling, and cloud data warehouses.

Core Technologies Mastered

Python · SQL · Statistics · Power BI / Tableau · scikit-learn · Cloud Warehouses

Program Overview

This 12–16 week, project-driven curriculum takes learners from foundational data wrangling and querying to advanced statistical reasoning, business intelligence, predictive modeling, and cloud-scale data engineering. Each module combines theory, hands-on exercises, and a mini-project, culminating in a portfolio-ready capstone project.

No	Module	Duration	Focus Area
1	Python & Data Wrangling	Week 1 – 3	Python
2	Advanced SQL & Data Modeling	Week 4 – 5	SQL
3	Statistics & Hypothesis Testing	Week 6 – 7	Statistics
4	Data Visualization & BI Tools	Week 8 – 9	Visualization
5	Predictive Modeling & Regression	Week 10 – 12	Machine Learning
6	Big Data & Cloud Data Warehouses	Week 13 – 14	Cloud / Big Data

Module 1: Python & Data Wrangling

Duration	Week 1 – 3
Outcome	Ability to clean, transform, and structure real-world datasets using Python, NumPy, and Pandas.

Topics Covered

- Python fundamentals: variables, data types, control flow, functions
- Data structures: lists, dictionaries, tuples, sets
- NumPy: arrays, vectorized operations, broadcasting
- Pandas: Series, DataFrames, indexing, filtering
- Data cleaning: missing values, duplicates, outliers
- Data transformation: merging, joining, reshaping (melt/pivot), groupby-aggregate
- Working with dates/times and text data (string operations, regex basics)
- Reading/writing data: CSV, Excel, JSON, SQL databases, APIs

Mini-Project: Clean and transform a messy real-world dataset (e.g., e-commerce or HR data).

Module 2: Advanced SQL & Data Modeling

Duration	Week 4 – 5
Outcome	Skill to design analytical data models and write advanced, performant SQL queries.

Topics Covered

- SQL refresher: SELECT, WHERE, GROUP BY, HAVING, ORDER BY
- Joins: inner, outer, self, cross joins
- Subqueries and Common Table Expressions (CTEs)
- Window functions: RANK, ROW_NUMBER, LAG/LEAD, running totals
- Query optimization and indexing basics
- Data modeling: normalization vs. denormalization
- Star schema vs. snowflake schema for analytics
- Designing fact and dimension tables
- Views, stored procedures, and basic transactions

Mini-Project: Design a data model and write analytical queries for a business dataset (e.g., retail sales).

Module 3: Statistics & Hypothesis Testing

Duration	Week 6 – 7
Outcome	Applied ability to reason about uncertainty and validate business decisions using statistical methods.

Topics Covered

- Descriptive statistics: mean, median, mode, variance, standard deviation
- Probability fundamentals: normal, binomial, Poisson distributions
- Sampling methods and the Central Limit Theorem
- Confidence intervals
- Hypothesis testing: null/alternative hypotheses, p-values, significance levels
- t-tests (one-sample, two-sample, paired)
- Chi-square tests for categorical data
- ANOVA for comparing multiple groups
- Correlation vs. causation
- A/B testing design and interpretation for business decisions

Mini-Project: Design and analyze an A/B test with statistical significance testing.

Module 4: Data Visualization & BI Tools

Duration	Week 8 – 9
Outcome	Capability to design compelling dashboards and communicate insights to business stakeholders.

Topics Covered

- Principles of effective data visualization and avoiding misleading visuals
- Python visualization: Matplotlib, Seaborn, Plotly
- Building interactive dashboards
- Introduction to BI tools: Power BI and/or Tableau
- Connecting BI tools to data sources
- Designing KPI dashboards for business stakeholders
- Storytelling with data: structuring insights for non-technical audiences

Mini-Project: Build an interactive dashboard answering a specific business question.

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Module 5: Predictive Modeling & Regression

Duration	Week 10 – 12
Outcome	Production-ready proficiency building, evaluating, and interpreting predictive machine learning models.

Topics Covered

- Introduction to machine learning: supervised vs. unsupervised learning
- Linear regression: assumptions, interpretation, evaluation metrics (R^2, RMSE, MAE)
- Logistic regression for classification
- Feature engineering and selection
- Train-test split and cross-validation
- Regularization: Ridge, Lasso
- Tree-based models: Decision Trees, Random Forests
- Model evaluation: confusion matrix, precision/recall, ROC-AUC
- Avoiding overfitting and the bias-variance tradeoff

Mini-Project: Build a predictive model (e.g., churn prediction or sales forecasting).

Module 6: Big Data & Cloud Data Warehouses

Duration	Week 13 – 14
Outcome	End-to-end capability to design and operate cloud-based data pipelines at scale.

Topics Covered

- Big data concepts: volume, velocity, variety
- Introduction to distributed computing (Spark fundamentals)
- Cloud data warehouses: Snowflake, BigQuery, Redshift (overview and comparison)
- Data pipelines and ETL/ELT concepts
- Batch vs. streaming data processing
- Working with cloud storage (S3/Blob/GCS basics)
- Cost and performance considerations in cloud analytics
- Data governance and security basics

Mini-Project: Capstone — ingest data into a cloud warehouse, transform it, and build a final analytics/ML deliverable combining skills from all modules.

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Suggested Tools & Stack

Category	Tools
Programming	Python (Jupyter / VS Code)
Database	PostgreSQL / MySQL
Visualization	Matplotlib, Seaborn, Power BI / Tableau
Machine Learning	scikit-learn
Big Data / Cloud	Spark, Snowflake / BigQuery, AWS / GCP / Azure basics

Assessment & Certification

- Weekly hands-on coding assignments and exercises aligned to each module
- Mini-project after every module (Modules 1–6)
- Mid-program checkpoint: end-to-end analysis combining Python, SQL, and statistics (Modules 1–3)
- Capstone project: full analytics-to-ML pipeline deployed on a cloud data warehouse
- Final proficiency evaluation covering data modeling, statistical rigor, and business communication

Upon successful completion, participants will demonstrate production-level analytics proficiency across the full data science lifecycle — from data wrangling and modeling through statistical validation, visualization, predictive modeling, and cloud-scale deployment.

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